Pigments

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Titanium Dioxide Pigment	IS 9788 – Titanium dioxide, rutile, for paints — SpecificationIS 33; IS 50	ISO 591-1 (Titanium Dioxide Pigments); ISO 787 Series (General Methods of Test for Pigments); ISO 787-25 (Optical Properties of Pigments),ISO 18451	ASTM D476; ASTM D2805 (Hiding Power of Paints); ASTM E308 (Spectrophotometric Colour Measurement); ASTM D2244 (Colour Difference); 0 (Quality Evaluation of Pigments), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280;	DIN EN ISO 787 series (General Methods of Test for Pigments and Extenders); DIN EN ISO 591-1 (Titanium Dioxide Pigments); DIN EN ISO 3262 series (Extenders); DIN EN ISO 11664 series (Colour Measurement); DIN EN ISO 787-5 (Oil Absorption)	REACH: Regulation (EC) No. 1907/2006 (titanium dioxide classification, dust exposure requirements, SVHC compliance); FDA: 21 CFR 73.575 (Titanium dioxide as colour additive where applicable)
Iron Oxide Pigment	IS 44:1991 – Iron Oxide Pigments for Paints,IS 33; IS 50	ISO 1248 (Iron Oxide Pigments); ISO 787 Series (Pigment Testing); ISO 787-25 (Colour Comparison),ISO 18451	ASTM D3723 (Pigment Dispersion); ASTM D2805 (Hiding Power); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); 0 (Pigment Testing), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280;	DIN EN ISO 1248 (Iron Oxide Pigments Specifications); DIN EN ISO 787 series (Pigment Testing); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 787-5 (Oil Absorption)	REACH: Regulation (EC) No. 1907/2006 (heavy metals, impurities, SVHC compliance); FDA: 21 CFR 73.3126 (Iron oxides as colour additives for specific uses)
Carbon Black Pigment	IS 40 – Carbon Black for Paints,IS 33; IS 50	ISO 1124 (Sampling of Carbon Black); ISO 5435 (Carbon Black Testing); ISO 787 Series (Pigment Testing),ISO 18451	ASTM D1510 (Iodine Number of Carbon Black); ASTM D1506 (Ash Content); ASTM D1514 (Sulfur Content); ASTM D2414 (Oil Absorption Number); ASTM D3265 (Tint Strength); ASTM D4821 (Carbon Black Dispersion), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series (Pigment Testing); DIN EN ISO 1124 (Carbon Black Sampling); DIN ISO 1306 (Carbon Black Rubber Grade); DIN EN ISO 787-5 (Oil Absorption); DIN EN ISO 787-11 (Volume and Apparent Density)	REACH: Regulation (EC) No. 1907/2006 (PAH impurities, SVHC compliance); FDA: 21 CFR 178.3297 (Colorants for polymers where applicable)
Chromium Oxide Pigment	IS 54:2025 – Green Oxide of Chromium for Paints — Specification,IS 33; IS 50	ISO 787 Series (General Methods of Test for Pigments); ISO 1248 (Inorganic Pigment Testing where applicable),ISO 18451	ASTM D2805 (Hiding Power); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); 0 (Pigment Evaluation), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664 series (Colour Measurement); DIN EN ISO 787-5 (Oil Absorption); DIN EN ISO 787-9 (Water-Soluble Matter)	REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (chromium compounds restrictions where applicable); FDA: 21 CFR 73.2750 (Chromium oxide greens as colour additive where permitted)
Zinc Oxide Pigment	IS 35 – Zinc Oxide for Paints,IS 33; IS 50	ISO 1248 (Pigment Testing); ISO 787 Series (General Methods of Test for Pigments),ISO 18451	ASTM D79 (Zinc Oxide Pigment Specification); ASTM D2805 (Hiding Power); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 9298 (Zinc Oxide Pigments); DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 787-5	REACH: Regulation (EC) No. 1907/2006 (zinc compounds); FDA: 21 CFR 73.1991 (Zinc oxide as colour additive where applicable)
Ultramarine Blue Pigment	IS 55:2025,IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement),ISO 18451	ASTM D2805 (Hiding Power); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); 0 (Pigment Testing), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664 (Colour); DIN EN ISO 787-5 (Oil Absorption); DIN EN ISO 787-9 (Water Soluble Matter)	REACH: Regulation (EC) No. 1907/2006 (sulfur compounds, impurities); FDA: 21 CFR 73.2725 (Ultramarine blue as colour additive where applicable)
Phthalocyanine Blue Pigment	IS 3574 (Part 2):2000 – Organic Pigments for Paints — Part 2: Phthalocyanines; IS 3574,IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement),ISO 18451	ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM D2805 (Hiding Power); 0 (Pigment Testing), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 787-5; DIN EN ISO 787-14 (Soluble Matter in Water)	REACH: Regulation (EC) No. 1907/2006 (copper compounds, impurities); FDA: 21 CFR 73.3126 where approved colour additive applications apply
Phthalocyanine Green Pigment	IS 3574 (Part 2):2000 – Organic	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour	ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement);	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 787-5	REACH: Regulation (EC) No. 1907/2006 (copper compounds,

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
	Pigments for Paints — Part 2: Phthalocyanines; IS 3574,IS 33; IS 50	Measurement),ISO 18451	ASTM D2805 (Hiding Power), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;		halogenated pigments); FDA: 21 CFR 73.3126 where applicable
Organic Pigment	IS 33; IS 50	ISO 787 Series (General Methods of Test for Pigments); ISO 7724 Series (Colour Measurement),ISO 18451	0 (Testing Pigments); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM D2805 (Hiding Power), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664 series; DIN EN ISO 787-5; DIN EN ISO 787-14	REACH: Regulation (EC) No. 1907/2006 (aromatic compounds, heavy metals, SVHC substances); FDA: only where approved food-contact colour additive applications exist
Inorganic Pigment	IS 33; IS 50	ISO 787 Series (General Methods of Test for Pigments); ISO 1248 (Inorganic Pigments where applicable),ISO 18451	0 (Pigment Testing); ASTM D2805 (Hiding Power); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664 series; DIN EN ISO 3262 series	REACH: Regulation (EC) No. 1907/2006 (heavy metals, impurities, SVHC compliance); FDA: applicable only for approved colour additives
Pearlescent Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement); ISO 2813 (Gloss Measurement),ISO 18451	ASTM E308 (Spectrophotometric Colour Measurement); ASTM D2244 (Colour Difference); ASTM D523 (Gloss Measurement); 0 (Pigment Testing), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 11664 series (Colour Measurement); DIN EN ISO 787 series; DIN EN ISO 2813 (Gloss where applied in coatings)	REACH: Regulation (EC) No. 1907/2006 (mica, metal oxide coatings, impurities); FDA: 21 CFR 73 series where approved colour additive use applies
Metallic Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement); ISO 2813 (Gloss),ISO 18451	ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM D523 (Gloss); 0 (Pigment Evaluation), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 2813	REACH: Regulation (EC) No. 1907/2006 (metal content, heavy metals, restricted substances); FDA: applicable only for approved food-contact colour applications
Aluminium Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 1248 (Inorganic Pigment Testing),ISO 18451	ASTM D2092 (Preparation of Zinc-Coated and Aluminium Surfaces where applicable); ASTM D2244 (Colour Measurement); ASTM E308 (Colour Measurement); 0 (Pigment Testing), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 1247 series (Aluminium Pigments); DIN EN ISO 787 series; DIN EN ISO 11664	REACH: Regulation (EC) No. 1907/2006 (aluminium compounds, metal powders); FDA: 21 CFR 178.3297 where used as polymer colourant
Bronze Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement),ISO 18451	ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); 0 (Pigment Evaluation), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 2813	REACH: Regulation (EC) No. 1907/2006 (copper/zinc compounds, heavy metal restrictions); FDA: only approved colour additive applications
Fluorescent Pigment	IS 33; IS 50	ISO 7724 Series (Colour Measurement); ISO 787 Series (Pigment Testing),ISO 18451	ASTM E991 (Color Measurement of Fluorescent Specimens); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 11664 series (Colour Measurement); DIN EN ISO 787 series; DIN EN ISO 16073 (Fluorescent Materials where applicable)	REACH: Regulation (EC) No. 1907/2006 (organic fluorescent dyes, restricted substances); FDA: only where specific colour additives are approved
Luminescent Pigment	IS 33; IS 50	ISO 17398 (Photoluminescent Safety Properties where applicable); ISO 787 Series (Pigment Testing),ISO 18451	ASTM E2073 (Photoluminescent Safety Marking Systems); ASTM E2030 (Photoluminescent Products); ASTM E308 (Colour Measurement), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 17398 (Safety Colours and Signs where applicable); DIN EN ISO 11664; DIN EN ISO 787 series	REACH: Regulation (EC) No. 1907/2006 (rare earth compounds, phosphors, heavy metals); FDA: only approved food-contact colour additive uses
Heat Resistant Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Stability	ASTM D2485 (Heat Resistance of Coatings); ASTM D2244 (Colour	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 16474	REACH: Regulation (EC) No. 1907/2006 (metal oxides, inorganic

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
		Testing),ISO 18451	Stability); ASTM E308 (Colour Measurement), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	(Weathering Evaluation in Coating Applications)	pigments, additives)
Weather Resistant Pigment	IS 33; IS 50	ISO 105-B02 (Colour Fastness to Artificial Light where textile-related); ISO 4892 Series (Artificial Weathering); ISO 787 Series,ISO 18451	ASTM G154 (UV Weathering); ASTM G155 (Xenon Arc Weathering); ASTM D2244 (Colour Change); ASTM E308 (Colour Measurement), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 16474 series; DIN EN ISO 4892 series; DIN EN ISO 11664; DIN EN ISO 787 series	REACH: Regulation (EC) No. 1907/2006 (UV stabilisers, pigments, restricted substances)
High Performance Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement); ISO 4892 Series (Weathering where applicable),ISO 18451	0 (Pigment Testing); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM G154 (Weathering Resistance), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 16474; DIN EN ISO 12944 (when used in protective coatings)	REACH: Regulation (EC) No. 1907/2006 (complex organic pigments, heavy metals, SVHC substances); FDA: only where approved colour additive use exists
Food Contact Pigment	IS 33; IS 50	ISO 1186 Series (Overall Migration Testing); ISO 787 Series (Pigment Testing),ISO 18451	ASTM D543 (Chemical Resistance of Plastics); ASTM D1899 (Migration Testing where applicable); ASTM D3359 (Adhesion where coating application applies), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN 1186 series (Food Contact Plastics); DIN EN 13130 series (Specific Migration); DIN EN ISO 787 series	FDA: 21 CFR 178.3297 (Colorants for polymers); FDA: 21 CFR 73/74 series (approved colour additives); REACH: Regulation (EC) No. 1907/2006 (SVHC and restricted substances)
Cosmetic Grade Pigment	IS 33; IS 50	ISO 22716 (Cosmetic GMP); ISO 787 Series (Pigment Testing),ISO 18451	0 (Pigment Testing); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); ASTM D5383 (Visual Colour Evaluation), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664; DIN EN ISO 10993 series where cosmetic-device contact applications apply	FDA: 21 CFR 73 and 74 series (Colour Additives Subject to Certification/Exemptions); REACH: Regulation (EC) No. 1907/2006
Ceramic Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 10545 Series (Ceramic Tile Testing where applicable),ISO 18451	ASTM C242 (Ceramic Whitewares Terminology); ASTM C1026 (Resistance to Freeze-Thaw where ceramic coating applies); ASTM E308 (Colour Measurement), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 10545 series (Ceramic Tile Testing where applicable); DIN EN ISO 11664 (Colour); DIN EN ISO 787 series	REACH: Regulation (EC) No. 1907/2006 (heavy metals, lead/cadmium restrictions); FDA: FDA 21 CFR 175.300 only for food-contact ceramic coating applications
Plastic Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 4892 Series (Weathering of Plastics); ISO 7724 Series (Colour Measurement),ISO 18451	ASTM D1729 (Visual Evaluation of Colour Differences); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); 0 (Pigment Testing), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 11664; DIN EN ISO 787 series; DIN EN ISO 4892 (Weathering of Pigmented Plastics)	REACH: Regulation (EC) No. 1907/2006 (heavy metals, restricted substances); FDA: 21 CFR 178.3297 (Colorants for polymers where applicable)
Rubber Pigment	IS 33; IS 50	ISO 787 Series (Pigment Testing); ISO 7724 Series (Colour Measurement),ISO 18451	ASTM D3677 (Rubber Identification where applicable); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement), ASTM D1208; ASTM D1210; ASTM D185; ASTM D280; ASTM D332;	DIN EN ISO 787 series; DIN EN ISO 11664; DIN ISO 188 (Rubber Accelerated Ageing where applicable)	REACH: Regulation (EC) No. 1907/2006 (PAHs, heavy metals, restricted substances)
Textile Pigment	IS 33; IS 50	ISO 105 Series (Textile Colour Fastness); ISO 787 Series (Pigment Testing),ISO 18451	ASTM D1729 (Visual Evaluation of Colour Differences); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); ASTM D5264 (Abrasion Resistance of Printed Materials), ASTM D1208; ASTM D1210;	DIN EN ISO 105 series (Textile Colour Fastness); DIN EN ISO 11664; DIN EN ISO 787 series	REACH: Regulation (EC) No. 1907/2006 (restricted azo dyes, SVHC substances); REACH Annex XVII (azo compounds restrictions)

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
			ASTM D185; ASTM D280; ASTM D332;		

Source field: Dyes and Pigments